Global Weather Report Dashboard

Power BI Project Report

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Project Title	Global Weather Report Dashboard
Tool Used	Microsoft Power BI Desktop (Power Query, DAX, Report View)

1. Dataset Description

The project uses weather data for five countries: **USA, UK, India, South Africa and Brazil**. Each country is stored as its own table in Power BI (*USA_Weather_data, UK_Weather_data, India_Weather_data, South_Africa_Weather_data, Brazil_Weather_data*). The records are daily observations and describe the weather of each state or region of the country across the months of the year.

Field	Description
State / Region	State of the country (e.g. Delhi, Gujarat, Karnataka in India; Bahia, Minas Gerais in Brazil; California, Florida in the USA)
Month / Day	Calendar month and day of the month (1-31) of each observation
Season	Spring, Summer, Autumn or Winter
Condition	Weather condition: Cloudy, Rainy, Windy, Thunderstorm, Humid, Clear, Sunny
Temperature	Temperature in °C
Wind_Speed	Wind speed in km/h
Humidity	Relative humidity in percent
AQI	Air Quality Index value
Outdoor_Activity	Suitability for outdoor activity: Good or Poor

2. Objectives

- Build an interactive, easy-to-read weather dashboard for five countries in Power BI.
- Summarise key weather indicators (temperature, wind speed, humidity and AQI) in one view per country.
- Analyse how weather conditions and seasons are distributed and how temperature varies day by day.
- Assess how suitable the weather is for outdoor activity in each country.
- Let the user explore the data through State and Month slicers and a Main Menu navigation page.

3. Data Cleaning Performed

Data preparation was carried out in Power Query before building the visuals:

- Imported each country dataset as a separate table and promoted the first row to column headers.

- Checked and set correct data types (text for State, Season and Condition; whole number for Day; decimal for Temperature, Wind_Speed, Humidity and AQI).
- Checked for null, blank and duplicate records and removed or corrected them.
- Standardised text values (consistent spelling of states, seasons and conditions) and trimmed extra spaces.
- Ensured Month names sort in calendar order so the Months slicer appears from January to December.
- Applied the same column structure to all five tables so that every country dashboard uses identical visuals.

4. Data Model

The model contains five fact tables, one per country, all with the same structure. Each country page is connected to its own table, so the slicers and visuals on that page filter only that country's data. Navigation between pages is handled through buttons: a **Home page** shows the five country flags as page-navigation buttons (CTRL+click to follow), and every country page has a **Main_menu** button to return home.

Page	Source table	Slicers	Visuals
Home	–	–	Flag navigation buttons
USA, UK, India, South Africa, Brazil	One table per country (e.g. India_Weather_data)	State, Months	4 KPI cards, donut, pie, stacked column, line chart

5. DAX Measures

The four KPI cards on each dashboard are calculated with average-based DAX measures (shown here for India; the same pattern is used for every country table):

```
Avg Temperature = AVERAGE ( India_Weather_data[Temperature] )
Avg Wind Speed  = AVERAGE ( India_Weather_data[Wind_Speed] )
Avg Humidity    = AVERAGE ( India_Weather_data[Humidity] )
Avg AQI         = AVERAGE ( India_Weather_data[AQI] )
```

The chart visuals use the Count of Condition (stacked by Season), the percentage share of Season and Outdoor_Activity, and Temperature plotted by Day and Season. All values respond to the State and Months slicers.

6. Dashboard Screenshots

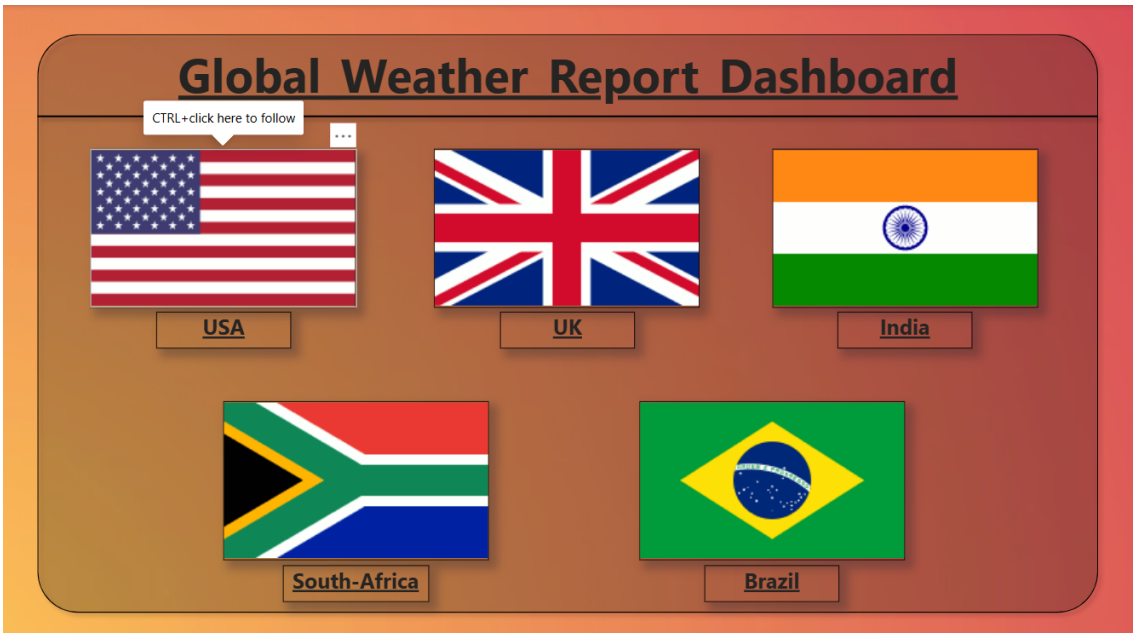


Figure 1: Home page – Global Weather Report Dashboard with navigation to the five countries

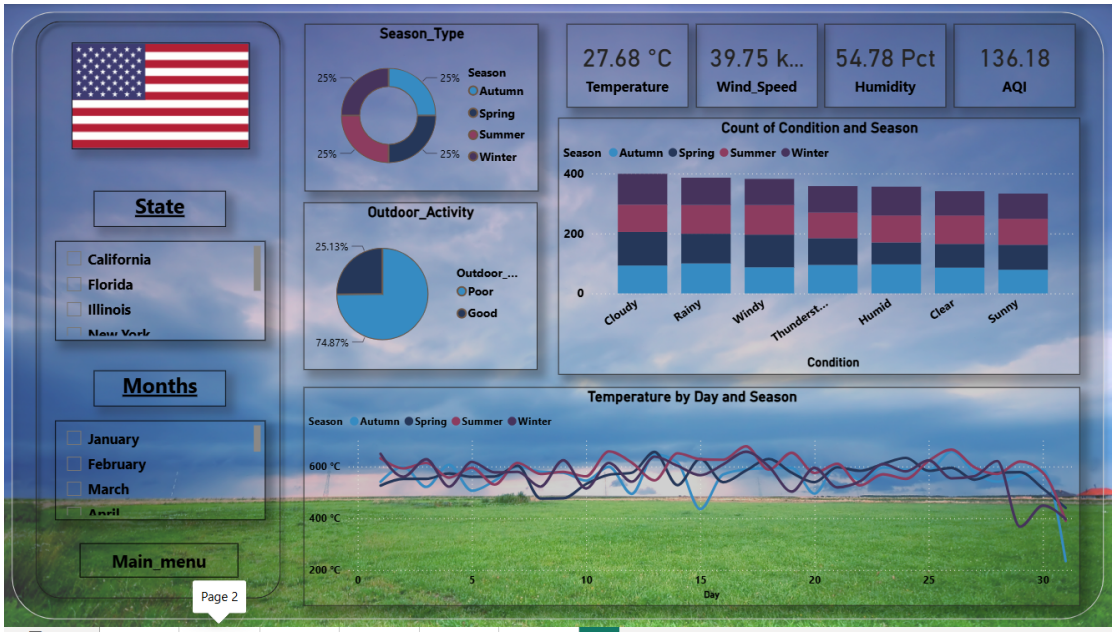


Figure 2: USA weather dashboard

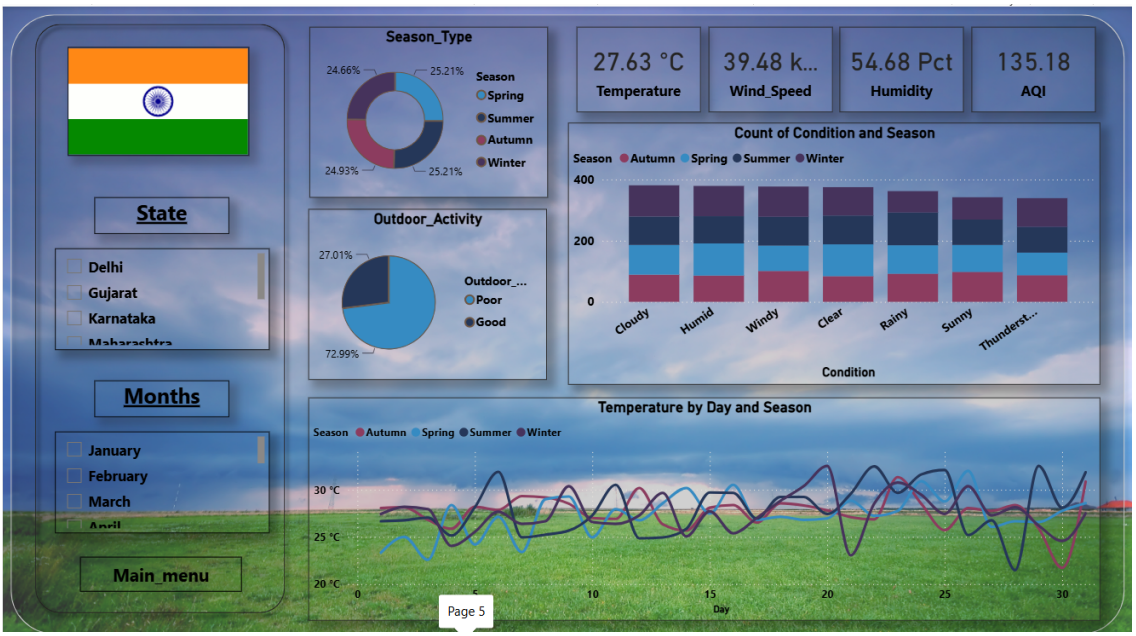


Figure 3: India weather dashboard

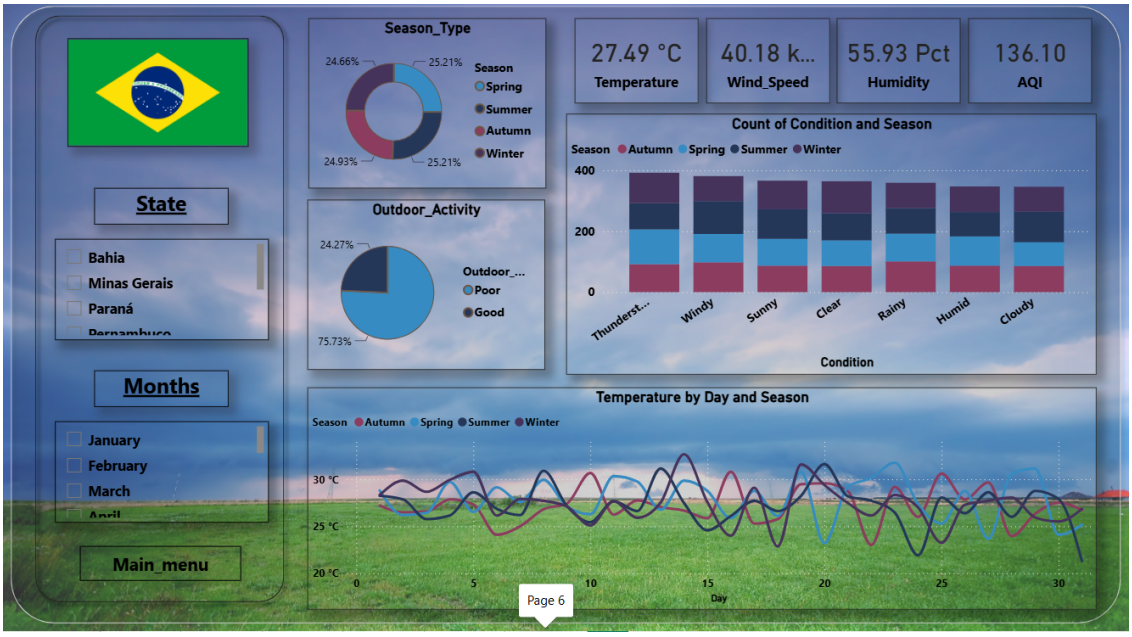


Figure 4: Brazil weather dashboard

7. Key Findings

The KPI cards and charts of the three country dashboards shown above give the following comparison:

Indicator	USA	India	Brazil
Avg Temperature	27.68 °C	27.63 °C	27.49 °C
Avg Wind Speed	39.75 km/h	39.48 km/h	40.18 km/h
Avg Humidity	54.78 %	54.68 %	55.93 %
Avg AQI	136.18	135.18	136.10
Outdoor activity Poor / Good	74.87 % / 25.13 %	72.99 % / 27.01 %	75.73 % / 24.27 %

- **Very similar averages:** temperature (about 27.5 °C), wind speed (about 40 km/h), humidity (about 55%) and AQI (about 135) are almost identical across the countries.
- **Balanced seasons:** Spring, Summer, Autumn and Winter each contribute roughly 25% of the records in every country.
- **Even condition spread:** all seven conditions occur with comparable frequency (roughly 340–400 records each), with no single condition dominating.
- **Outdoor activity is mostly poor:** about 73–76% of days are rated Poor. India has the best share of Good days (27.01%) and Brazil the lowest (24.27%).
- **Air quality concern:** an AQI near 135 falls in the 101–150 band, which is considered unhealthy for sensitive groups.
- **Temperature trend:** in India and Brazil the daily temperature moves mostly between about 22 °C and 32 °C across the month with frequent day-to-day fluctuation in all seasons.

8. Conclusion

The Global Weather Report Dashboard successfully brings weather data of five countries into a single, interactive Power BI report. With the Main Menu navigation, State and Month slicers, KPI cards and season/condition charts, a user can quickly explore and compare temperature, wind, humidity, air quality and outdoor-activity suitability. The analysis shows that the countries have very similar overall weather profiles, that seasons and conditions are evenly distributed, and that a large majority of days are rated poor for outdoor activity with moderately elevated AQI. The project demonstrates data cleaning, data modelling, DAX measures and effective dashboard design, and the same layout can be extended with more countries or additional measures in future.